

# GENSET

## 320 kVA LIQUID (RADIATOR) COOLED

### INSTALLATION GUIDELINE & OWNER'S MANUAL



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## Overall Specifications of KOEL Green Liquid ( Radiator ) cooled Silent Gensets

| Genset Model | Engine Model | Engine Power at 1500 rpm. kW (hp) | Prime Power Genset rating at 0.8 pf (kVA) | Overall Canopy Dimensions L x W x H (mm) Tol ± 50 | Approx.wet weight ( kg) | Fuel Tank Capacity(Ltr) |
|--------------|--------------|-----------------------------------|-------------------------------------------|---------------------------------------------------|-------------------------|-------------------------|
| KG4-320WS1   | 6SL90ETA 4G3 | 279.5(380 )                       | 320                                       | 4750 x 1700 x 2575*                               | 4200                    | 600                     |

\* Including silencer

- Genset ratings are as per ISO 8528 performance class G3.
- Prime Power Rating is the maximum power available continuously for a variable electrical load for unlimited number of hours per year under standard operating conditions.
- Specific gravity of diesel is considered as 0.845 (+5 % tolerance as per ISO 3046) .

### Sizing of KGPI Genset

To select the right generating set under a given set of circumstances, knowledge of generating set and its performance under various conditions is necessary. It is also equally necessary to understand load characteristics and to make a load analysis. Beyond that, the effect of the voltage regulation systems plays an important role.

It is better to select a slightly larger generator capacity than the available load; the reason is that additional uses are invariably observed after it has been installed. These are usually not predictable. By having a margin available it is possible to power some loads which may have been overlooked during Genset selection, or which may be added after the installation of the Genset

### Ratings

The rating of an alternator is expressed either in kilowatt (kW) or kVA. The relation between kW and kVA rating is :  
 $kW = kVA \times \text{Power Factor}$ .

$$kW = 1.732 \times V \times A \times \text{Power Factor (for 3 phase alternator)} / 1000$$

$$kW = V \times A \times \text{Power Factor (for a single phase alternator)} / 1000$$

Where V = Line voltage; A = Line current

Load Variation between phase should not more than 10% and min 50% load should be available at the time of commissioning

### Power Factor

The power factor of electrical system depends upon the nature of characteristics of load e.g Induction motors, Furnaces, thyristor, power electronics etc. Power Factor is lagging for inductive load. Power Factor is leading for capacitive load. Power factor is unity for purely resistive loads. Normally, a generator is designed for a power factor of 0.8 The table below gives power factors of non-motor loads

| Type of load                                       | Power Factor |
|----------------------------------------------------|--------------|
| Incandescent lights                                | 1.00         |
| Heating elements, ovens                            | 1.00         |
| Fluorescent lights                                 | 0.90         |
| Induction furnaces                                 | 0.60 - 0.70  |
| Arc furnaces                                       | 0.80 - 0.90  |
| Welding Set Transformer)                           | 0.60         |
| Welding Set( AC Motor-DC Generator) & Transformers | 0.80 - 0.90  |

- Load variation between phases should not be more than 10%.
- $KVA \times 1.39 = \text{AMP per Phase for III phase gen set}$  &  $KVA \times 1.39 \times 3 = \text{AMP for I phase gen set at 100\% load}$  .  
 Minimum 50% of AMPs per phase is required at the time of commissioning.

### Deration Factor

The engine's horse power as mentioned on its name plate is the HP which an engine can deliver in MSL at NTP conditions. HP delivered at site will vary depending on site conditions (deration due to altitude , temperature and humidity ) for details of site derations please consult seller or KOEL area office.

Input HP (Metric ) = . kW / 0.736 x Alternator Efficiency.

## Load Calculation

Steps to be followed while sizing the Genset.

### Step 1:

Collect all necessary basic data of load. Tabulate the details such as lighting load, motor load, A/C load, Heaters load etc with quantity and name plate details.

For motor loads, mention starting method also, i.e. Type of starter used.

### Step 2 :

Applicable For Three Phase Genset Convert all single phase loads to three phase loads by distributing equally on three phases.

### Step 3:

Starting power (kVA) of each equipment is based on type of starting method.

For eg. In case of motors,

Starting by DOL will pull up to 6 to 7 times the rated full load kW of a motor.

Star/Delta will pull up to 2 to 3 times the full load kW of the motor.

Soft Start will pull up to 1.2 to 1.5 times the full load kW of the motor. Rectifier loads may also pull up to 50% above their rated KVA.

### Step 4:

Sum up all the steady state running loads in kVA and sum up all the starting kVA's . These two figures will be used to size up the Genset and sequence of loading.

### Step 5:

Determine the maximum allowable voltage dip which the Genset should experience when the total starting KVA is applied on this Genset. This voltage dip percentage is usually known to the customer based on equipment type to be powered.

### Step 6:

The previously obtained figures (Running KVA, Starting KVA, Voltage, Voltage Dip, Loading sequence) will decide Genset size.

## Installing KGPL Genset

### Location



Dust and Fumes are the greatest danger to generating set as they could lead to clogging of cooling and air system which may affect Genset performance.

- If KGPL Genset is to be installed in Free field condition, please ensure dust free location and clear space of 2 times the height of Genset.
- If KGPL Genset is to be installed in the room, please ensure proper cross ventilation, clear space of minimum 2 meters from all sides.
- If KGPL Genset is to be installed on a roof top, please ensure structural clearance, proper cross ventilation with respect to wind direction, clear space of minimum 2 meters from all sides
- If KGPL Genset is to be installed in the basement, please ensure ventilation with respect to Air requirement and clear space for easy maintenance.

### Layout

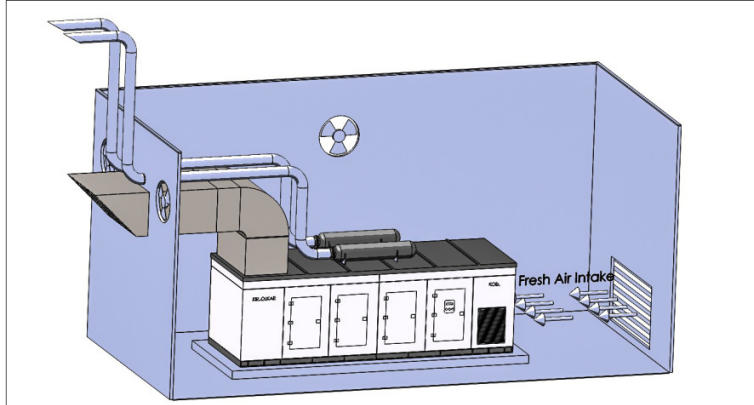
- Allow easy accessibility to Genset for easy operation and maintenance.
- Locate the Genset considering the prevailing wind direction.
- Genset installation should be as close as possible to the load. However in some cases , like stone crusher plant, Garment plant, Cement factory, Paint factory, chemical factory, etc the Genset should be isolated from the dust, hazardous and chemical fumes contained in the surrounding air.
- In case, Genset is to be installed in heavy dusty conditions, it is recommended to use heavy duty air cleaners (Please contact Seller or KOEL Area office for further guidelines).
- For fuel supply and fuel storage arrangement for Genset , follow the local rules and regulations.  
\* The environment protections . \* The fire protection .
- In case of high humid atmospheric conditions, an anti condensation heater should be used for the alternator.
- A rooftop installation requires proper planning and the structural design considerations for dynamic loading.
- Proper consideration should be given to take exhaust gases away from Genset location and shielded for hot surface.

## Noise

- There should be sufficient space around the Genset to avoid resonance and echo effect which contributes to abnormal sound from Genset.
- The space should be open without any obstacles.

## Ventilation

- Cross ventilation and free flow of cool, clean and fresh air
- The space should be open without any obstacles.
- For room and basement installations, supply of fresh air and removal of hot air should be done through forced ventilation with suitable air ducts.
- Maximum allowed temperature rise above ambient in Genset Enclosure at air intake of engine is 7–9 °C



## Foundation

Foundation is one of the important factors affecting the successful operation and life of a generating set. Improper foundation may result in alignment and vibration problems which may subsequently lead to pre-mature failure of Genset Components and Safety.

The basic functions of foundation are given below.

- To provide leveled platform to seat and seal the Genset.
- To support the weight of the entire generating set.
- Support the dynamic load of Genset while running.
- Absorb the vibration produced by the rotating and reciprocating masses.
- Isolate the surrounding structures by absorbing the vibration of generating set while running.

**320KVA DG Foundation Sizes :- 5350(L) x 2300(W) x 200(H)mm ( Above GND 200 mm & Below GND depend upon soil condition)**

While casting the concrete foundation, the bearing strength of the soil at site of installation should be determined. The table below gives approximate safe bearing capacity of various soils for ref.

- The length and width of foundation should be at least 300 mm more on each side than acoustic enclosure length and width respectively.
- It is recommended to have foundation height of 150 to 200 mm above ground level to maintain cleanliness and avoid flooding.
- Check the foundation level diagonally as well as across the length and width for even flatness and same should be within  $\pm 50$  mm in horizontal plane in any one direction only.
- Ensure that the concrete is completely cured before positioning the enclosure.
- Ensure that the foundation to support 1.5 times of the total wet weight of the single generating set installation and 2 times of the total wet weight for the multiple generating set installation.
- For rooftop installations, it is necessary to have planning and structural design approval.



Do not install acoustic enclosure on loose sand or clay. Avoid hard / sharp projections such as stone or steel parts on foundation surface, which may damage fuel tank at the bottom.

Avoid uneven foundation which may lead to improper resting of Genset on foundation, thereby leading to leakage of sound and vibration on Genset.

If Genset is to be mounted on foundation having uneven surface under unavoidable circumstances, use appropriate thickness of rubber matting above foundation for positive sealing of Genset with base.

## Unloading and Positioning

- All the Gensets are provided with suitable lifting arrangements. unload the Genset, by lifting with proper lifting tackle/nylon sling, /soft sling with spreader bar/spacer plate of suitable capacity to avoid any damage
- Do not drag the Genset on floor/ground.
- Use lift points on the acoustic enclosure (canopy) which provides support to the equipment when it is moved.
- Don't use single belt for DG set shifting.



## Exhaust System

Genset Exhaust system directs the hot flue gases into the atmosphere harmlessly.

On KGPL Genset, exhaust silencer and expansion bellow are the integral parts of the exhaust system. At the time of Genset installation, please ensure proper alignment of exhaust silencer with respect to expansion bellow.

While taking exhaust gases away from Genset above stack height, please refer to the table below for selecting the size of exhaust pipe.

As per Statutory regulations, exhaust pipe has to be extended above the surrounding stack as below.

$$H = h + 0.2 \times \sqrt{\text{kVA}}$$

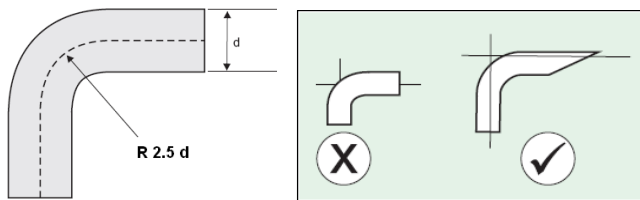
Where H = Minimum height of exhaust stack, h = height of building.

Note :

- If the extension of exhaust pipe is more than 5 meters above Genset, increase pipe diameter by 13 mm for every 3 meters.

Exhaust bend :

Bends used in exhaust piping should have a long smooth bend as shown in figure to ensure smooth flow of exhaust gases.



Note :

- If No. of bends are more than 4 , increase the diameter of exhaust pipe and bend by 13 mm for addition of each bend .
- The exhaust outlet should be in the direction of prevailing wind.
- Exhaust outlet should have 30° cut at the tail end Coarse wire mesh.

## Rain Cap

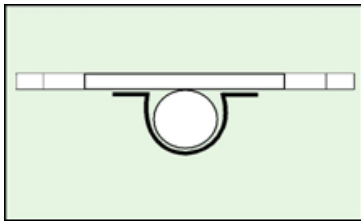
If horizontal exhaust outlet as recommended above is not possible , provide a rain cap over exhaust pipe end as per the figure below . The distance between exhaust pipe and rain cap should be higher than diameter of pipe ( minimum 2.5 D) with coarse wire mesh.

It is also recommended that horizontal run of exhaust piping should slope downwards away from engine to the condensate trap . Silencer should be installed with drain plug at bottom.



## Exhaust Support

Proper support is required for the extended exhaust pipe above Genset to ensure no self weight is transferred on to the Genset



## Type of Exhaust Pipe

Use black MS ERW (Electric Resistance Welded) - Class B Type pipes. Galvanized Iron pipes should not be used. Exhaust pipes should be joined through flanges.

Exhaust pipe throughout should be cladded by mineral wool with aluminum sheet 24 SWG Minimum thickness of cladding should be 50 mm.

## Liquid Cooled Engine

Liquid Cooled Engines use premixed coolant for removal of heat generated in the engine  
Due to internal combustion & to maintain optimum operating temperature for efficient engine operation.

## Lub Oil System

- KOEL engines utilize the pressure lubrication system with the lub oil filter adequately designed to handle the flow of lubricating oil at the designed pressure and temperature for Lubrication, Cooling and Cleaning of internal components. KGPL Gensets are delivered with initial fill of Lub Oil "K-Oil Super".
- Check the Lub Oil level with the help of Dipstick to "H" Mark and top up if required with "K-Oil Super" only. K-Oil Super SAE 15W 40 is a specially blended engine oil which ensures :
  - Increased change intervals.
  - Better oxidation stability and hence less top-ups.
  - Reduced frictional wear.
  - Increased internal cooling.
  - Protection against rust and corrosion.
  - Reduced carbon deposits.

## Lub Oil System

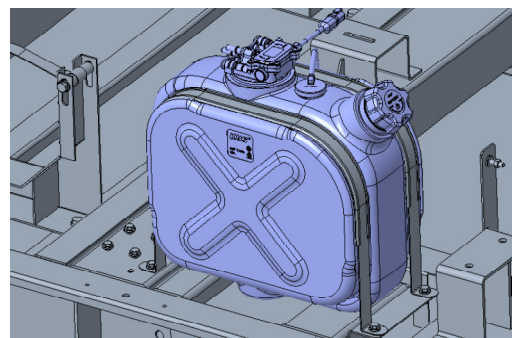


Use of the wrong grade of oil could lead to:

- Overheating of the engine.
- Sluggish performance.
- Excessive fuel consumption
- Increased wear of bearings and other moving components.

## Ad Blue Specifications:

- Use & handling of Ad Blue solution in Kirloskar Genset.
- Engines as per the requirement of CPCB (Central Pollution Control Board) Stage IV+ adopting the SCR technology (for NOx reduction) on Product.
- When used and handled according to specifications, the product does not have any Harmful effects according to our experience and the information provided to us.
- **Arrangement of Ad Blue tank 45 Ltr 1 No. tank arrangement.**



## Fuel System

- The fuel (High Speed Diesel) should comply with the International Standards. DIN 51601 , IS : 1460 1995 , ASTM D 975-81: 1D & 2D
- Ensure fuel filling point is accessible and clean all the time.
- Ensure all fuel piping is leak free.
- Avoid fuel spillage
- Ensure fuel dilution does not occur.
- Care should be taken while filling of tank to prevent ingress of dust & moisture inside diesel tank.
- It is recommended to fill fuel after stopping of the Genset to avoid condensation .



It is recommended not to fill fuel during running of Genset

## Electrical System

### Earthing

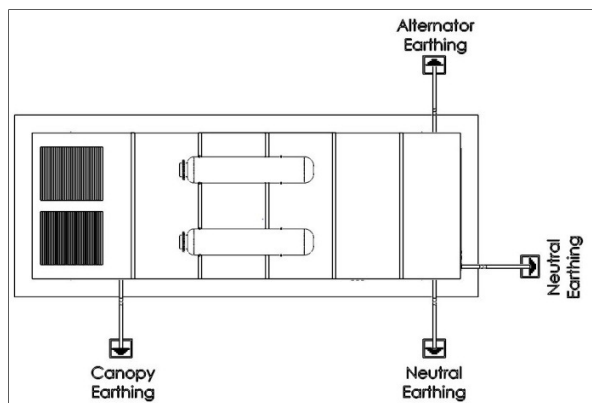
Any leakages in current will be earthed through the shortest route in the link.

KGPI Genset should be connected to earth in accordance with local regulations.

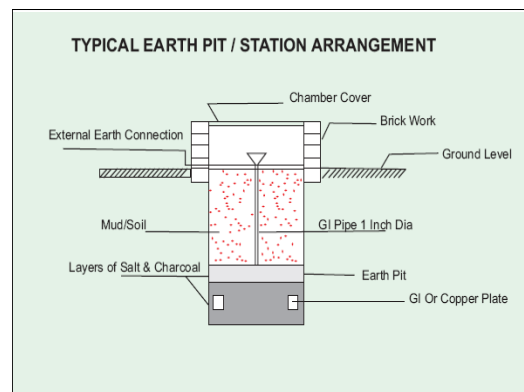
The size of the link used for the main earth connection should be of adequate size as shown in table below.

The generating set and all associated equipment must be earthed before the set is put into operation. 4 numbers of Earth pits are required as per Indian Electricity rules / local electricity regulations.

- 2 Earthing pits for Genset / control panel body
- 2 Earthing pits for neutral.
- Distance between 2 earth pits should be more than 2 meter.
- Every Earthing Resistance should be below  $5\Omega$ .
- Earthing pit Depth minimum 10ft.



Typical Earth Pit diagram is as shown in figure



| Genset Rating | Recommended Earth Cable / Strip |
|---------------|---------------------------------|
| 320 kVA       | 8 SWG – Copper / 25 x 3 mm GI   |

### Power Cable:

Cabling from Alternator to Panel should be done with proper size of cable and lugs.

Typical cable sizes provided in the table are indicative . Please refer to cable manufactures for details.

- Ensure tight crimping / terminations to avoid overheating and burning of the cable and terminals.
- Ensure correct size of gland & gland plates are used to avoid tension on alternator terminal , terminal box and control panel terminations.
- Ensure phase colour coding is adopted while terminating cables at both the ends of alternator and control panel.
- Ensure proper bending radius is given to the cable to avoid excessive tension on cable termination. For extended control cabling use 10/12 core 2.5 sq.mm. of armoured copper cable.
- Any type of control & monitoring cables should be sheathed and should be within 5 meters length of accurate performance.
- Ensure that the weight of cables does not rest on alternator terminations . Proper support to be provided from the ground separately through cable tray.

### Typical Cable Sizes for KGPI Genset

Cable selection chart:

| 3-Phase Genset |                    |                      |              |
|----------------|--------------------|----------------------|--------------|
| Genset Rating  | Amp Rating @0.8 pf | Cable Sizes in Sq mm |              |
|                |                    | Copper Multistrand   | Aluminium    |
| 320 kVA        | 445                | 150 X 2runs          | 240 x 2 runs |

#### Cabling- ISI Mark :

|        |                                      |
|--------|--------------------------------------|
| 320kVA | 3.5 core x150 sq.mm- 3 Run Aluminium |
|--------|--------------------------------------|

#### Important Notes:

1. Please use proper size cable glands.
2. Type of cable gland: Single/double Compression Heavy Duty Cable Gland.
3. Material of cable gland: Brass/Aluminum
4. The amp. rating of cable is considered as per duct laying method and 700 C operating Temperature mentioned in manufacturers catalogue.
5. Considered amb temp: 40° C.
6. For grouping of cables (multiple runs) the derating factors are considered as per duct laying method of cable.
7. The distance between two parallel cables is considered 30 mm for duration.

#### Battery

| Engine model | Battery Specification | Battery cable size |
|--------------|-----------------------|--------------------|
| 6SL90ETA 4G3 | 12 V x 180 Ah- 2 nos  | 95 sq.mm           |

Ensure that the polarity of the battery connections are correct before switching on the DC controls.  
Never attempt to start the Genset if battery charge is insufficient to start the engine.  
Get the battery charged or replaced and then try to start.  
During installation and maintenance, disconnect the battery.  
For prolonged storage, used static battery charger to keep battery fully charged.

Specific Gravity of Lead Acid in the Battery at various stages:

- 100% Charge – 1.27 to 1.24 gm/cc (Yellow)
- 70% Charged – 1.24 to 1.20 gm/cc (Blue)
- Needs Recharge – Below 1.20 gm/cc (Red)

#### Starting and Stopping A New KGPL Genset

##### Lubrication:

- Check Lube oil level with help of Dipstick and if required, top up to H-Mark with “K-Oil Premium”.

##### Fuel:

- Ensure fuel tank is clean from inside.
- Fill fuel tank with clean High Speed Diesel.
- Bleed the fuel system as shown in maintenance manual.



## Cooling:

### Liquid Cooled Engines:

- Check Coolant level inside Radiator and compensatory Tank. If required Top up with "K-Oil Premium " .

## Electrical:

- Check Battery Polarity.
- Check Battery cables are correctly connected and secured.
- Check all other electrical connections and tightness.
- Ensure Emergency Push Button is in released condition.
- Ensure Load Change Over Switch / Breaker is in Off condition.

## Procedure: Starting KGPL Genset

### Before Starting

- Clean the Genset with dry cloth.
- Check the lube oil, coolant and fuel levels – top up if required as per recommendation given.
- Check for V-belt, Hoses and external leakage if any.
- Ensure Load Change Over Switch / Breaker is in Off condition.

### On Starting

#### Starting Genset

- Do not crank engine for more than 5 seconds.
- As soon as engine fires, release the push button .
- If engine fails to start, wait for a minute and try again.
- Check for leakages if any.
- Allow the engine to idle for 3 minutes and then start loading the Genset gradually in steps.
- Check for Engine parameters like Oil Pressure and Battery charging rate at regular intervals while the Genset is in operation and maintain log book.
- Check for exhaust smoke.
- Check for abnormal noise if any.
- Check for A C Alternator parameters like Voltage, Ampere, KW, PF on all phases.
- Ensure load is balanced in all phases as per recommendation and within rated output of Genset.
- Check all gauges and meters provided in Control Panel are working.

## Procedure: Stopping KGPL Genset

### Before stopping

- Remove the load from Genset and let it run for about 3 minutes.

### After Stopping

- Check for leakages if any.
- Top up the Fuel Tank.
- Ensure proper closing of Genset Enclosure

### LOG SHEET (Sample for use)

The user is requested to maintain a separate log book registering the following mentioned parameters in the table. Following tables are meant for providing a reference to you .

|                                                |                                |  |  |  |  |
|------------------------------------------------|--------------------------------|--|--|--|--|
| Date                                           |                                |  |  |  |  |
| Daily use in hrs.                              |                                |  |  |  |  |
| Hour meter reading                             |                                |  |  |  |  |
| Cooling Water Temp (°C)                        |                                |  |  |  |  |
| Lub. Oil                                       | Pressure (Kg/cm <sup>2</sup> ) |  |  |  |  |
|                                                | Temp (°C)                      |  |  |  |  |
| Fuel added (ltr.)                              |                                |  |  |  |  |
| Lub. Oil added                                 |                                |  |  |  |  |
| Engine (rpm)                                   |                                |  |  |  |  |
| Current (Amp) 3-phase                          |                                |  |  |  |  |
| Voltage                                        |                                |  |  |  |  |
| Frequency (Hz)                                 |                                |  |  |  |  |
| PF & kW if provided                            |                                |  |  |  |  |
| kWH meter reading if provided                  |                                |  |  |  |  |
| Remarks – Record events of maintenance/repairs |                                |  |  |  |  |

### Maintenance Tips

First oil change @ 50 hrs or 1 month from the date of commissioning.

| Series | Oil Change | Oil Filter   | Fuel Filter   | Air Cleaner element                     | V Belt Change |
|--------|------------|--------------|---------------|-----------------------------------------|---------------|
|        | hours      | Change hours | Change hours* | change hours**                          | hours         |
| SL90   | 500        | 500          | 500 (1.1 Ltr) | 1000- for dry type air cleaner element. | 1000          |

\* Fuel Filter Pre and Micro,oil filter to be replaced alternatively after 500 hrs.

\*\* Air filter change is based on Ideal site condition, however extreme dusty condition change frequency is based on occurrence of red band in restriction indicator. It is recommended to get the routine maintenance done through KOEL Authorized dealer.

### Cleaning and Preservation of KGPL Genset

- When the Genset is kept idle for more than 6 months, it is advisable to protect it from atmospheric moisture, condensates, internal corrosions and rodents.

### Warranty:-

Kirloskar Green DG sets comes with the warranty of 2 years from the date of installation/commissioning or 5000 operating hours or 30 calendar months from date of dispatch, whichever is earlier. Warranty is subject to use of K-Oil Premium, genuine filters, K-Cool Super Plus coolant and services sourced through Kirloskar Green authorized Service Dealer of the area.

### Customer Support Network

- The after sales service to KGPL Genset is provided through KOEL's country-wide network of authorized spare and service dealers. Service dealers are equipped with all necessary tools and tackles to take care of every need of service.
- We understand how important after-sales support is to your business. Where your business operations are far reaching across the country & moreover when they are located in absolute remote places, your business counts on reliability & availability of prompt service to ensure minimum downtime.
- We are committed to provide adequate after sales service support & believe in creating an enduring relationship with every individual customer we serve.
- Our service team is always round the corner to provide onsite assistance and comprehensive Service support ensuring your power needs are fulfilled
- Before commissioning the KGPL Genset, please go through the contents of the warranty document.
- For any of your service requirements please contact nearest service dealer or KALA Help Desk .
- To register a complaint Engine Sr. No mandatory.

#### Registering complaint on Below **Contacts** :

Email ID : [aw@kalabiz.com](mailto:aw@kalabiz.com) / [sdp@kalabiz.com](mailto:sdp@kalabiz.com)  
Mobile No. : [+91 9545556225](tel:+919545556225) / [+91 9545556320](tel:+919545556320)

### Kala Genset Pvt Ltd also provide installation services at site.

- In view of continuous product updating and design changes, all above specifications and dimensions are subject to change without prior notice. Also refer to latest product catalogue for updates.
- For the site conditions other than standard operating conditions, consult KALA or KOEL's Area Office.